

# Assignment No : 08

## 1. Title

Case Study on Kernel-based Virtual Machine (KVM): Installation and Configuration of Virtualization

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## 2. Problem Statement

Kernel-based Virtual Machine (KVM) is a virtualization technology built into the Linux kernel that allows users to run multiple virtual machines on a single physical system.

The objective of this case study is to install and configure KVM virtualization on a Linux system and create a virtual machine.

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## 3. Objectives

- To understand the concept of **virtualization**
  - To study **KVM (Kernel-based Virtual Machine)**
  - To install and configure KVM on Linux
  - To create and manage virtual machines
  - To gain practical knowledge of virtualization tools
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## 4. Introduction to KVM

Kernel-based Virtual Machine (KVM) is an open-source virtualization technology built into Linux.

It converts the Linux kernel into a hypervisor, allowing multiple operating systems to run simultaneously.

KVM works with tools like:

- QEMU (for hardware emulation)
- libvirt (for VM management)
- virt-manager (graphical interface)

### What is Virtualization?

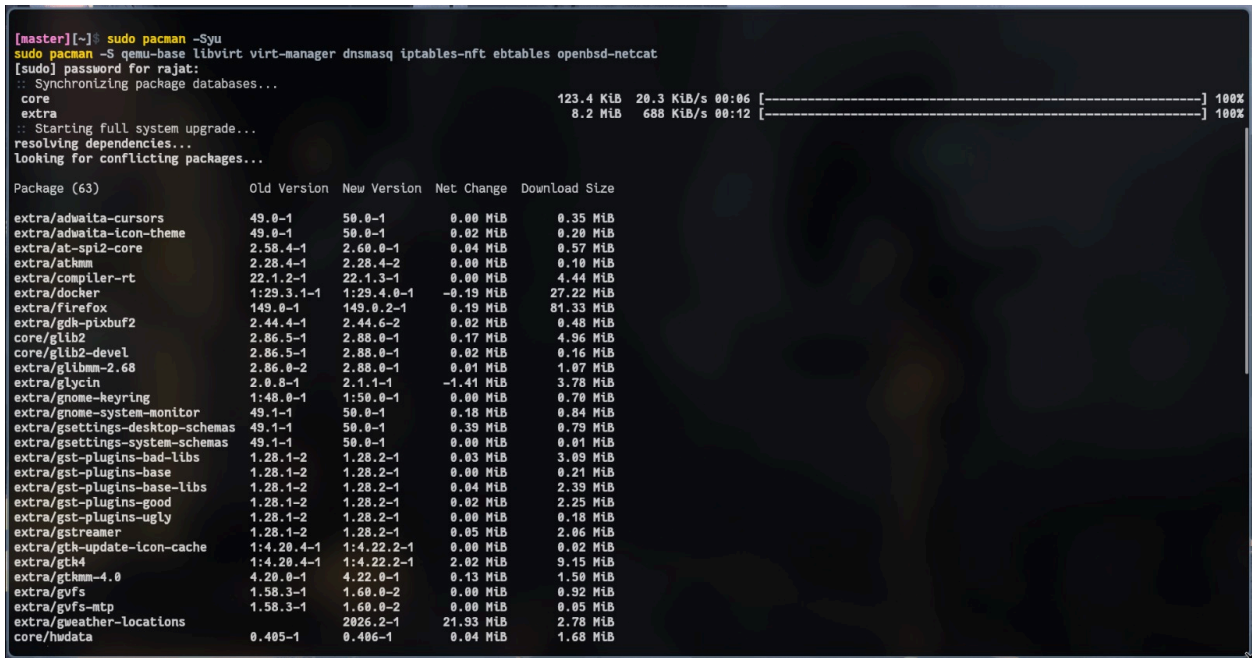
Virtualization is the process of creating virtual versions of physical resources such as servers, storage, or operating systems.

# Installation and Configuration of virtualization using KVM.

## COMMANDS FOR KVM INSTALLATION & VM SETUP

### 1. System Update

sudo pacman -Syu

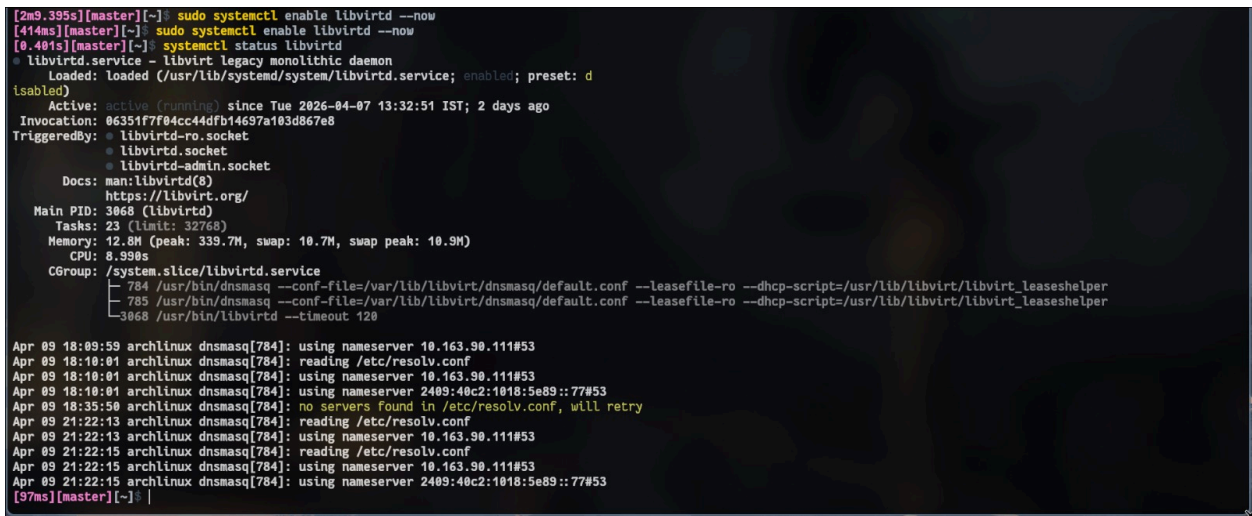


### 2. Install Required Packages

sudo pacman -S qemu-base libvirt virt-manager dnsmasq iptables-nft ebtables openbsd-netcat

### 3. Enable libvirt Service

sudo systemctl enable libvirtd --now  
systemctl status libvirtd



### 4. Add User to Required Groups

sudo usermod -aG libvirt \$(whoami)  
sudo usermod -aG kvm \$(whoami)

### 5. Install Kernel & Headers

sudo pacman -S linux linux-headers  
sudo reboot

6. Check Bridge Module

lsmod | grep bridge

```
[97ms][master][~]$ lsmod | grep bridge
bridge                458752  1 br_netfilter
stp                    12288   1 bridge
llc                    16384   2 bridge,stp
[55ms][master][~]$ |
```

7.Create Default Network

sudo nano /etc/libvirt/qemu/networks/default.xml

```
GNU nano 8.7.1 /etc/libvirt/qemu/networks/default.xml
!
WARNING: THIS IS AN AUTO-GENERATED FILE. CHANGES TO IT ARE LIKELY TO BE
OVERWRITTEN AND LOST. Changes to this xml configuration should be made using:
  virsh net-edit default
or other application using the libvirt API.
→
<network>
  <name>default</name>
  <uuid>628fc689-65c4-4a35-affd-339ed4a638d9</uuid>
  <forward mode='nat' />
  <bridge name='virbr0' stp='on' delay='0' />
  <mac address='52:54:00:87:1c:3c' />
  <ip address='192.168.122.1' netmask='255.255.255.0'>
    <dhcp>
      <range start='192.168.122.2' end='192.168.122.254' />
    </dhcp>
  </ip>
</network>
```

```
<network>
  <name>default</name>
  <forward mode='nat' />
  <bridge name='virbr0' stp='on' delay='0' />
  <ip address='192.168.122.1' netmask='255.255.255.0'>
    <dhcp>
      <range start='192.168.122.2' end='192.168.122.254' />
    </dhcp>
  </ip>
</network>
```

8.Define, Start & Enable Network

sudo virsh net-define /etc/libvirt/qemu/networks/default.xml  
sudo virsh net-start default  
sudo virsh net-autostart default

9. Verify Network

ip a | grep virbr0

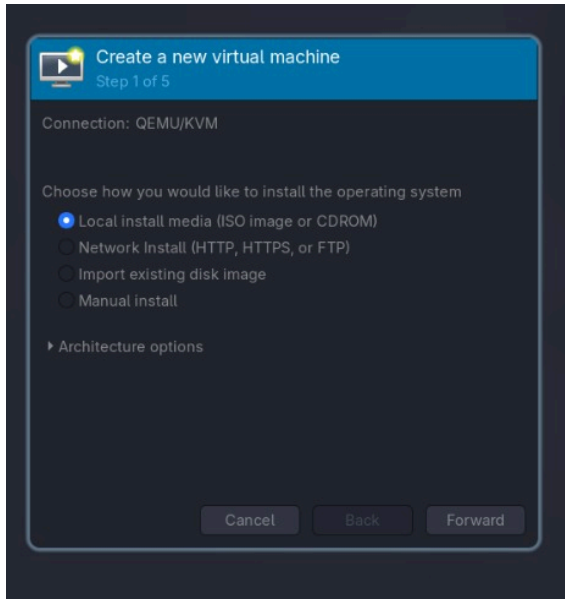
10.Launch Virtual Machine Manager

virt-manager

VM CREATION STEPS (INSIDE VIRTUAL MACHINE MANAGER)

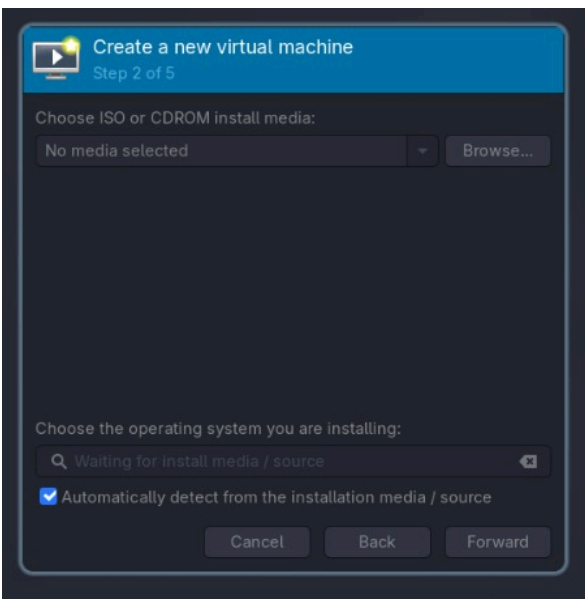
Step 1: Start VM Creation

Click "Create a new virtual machine"



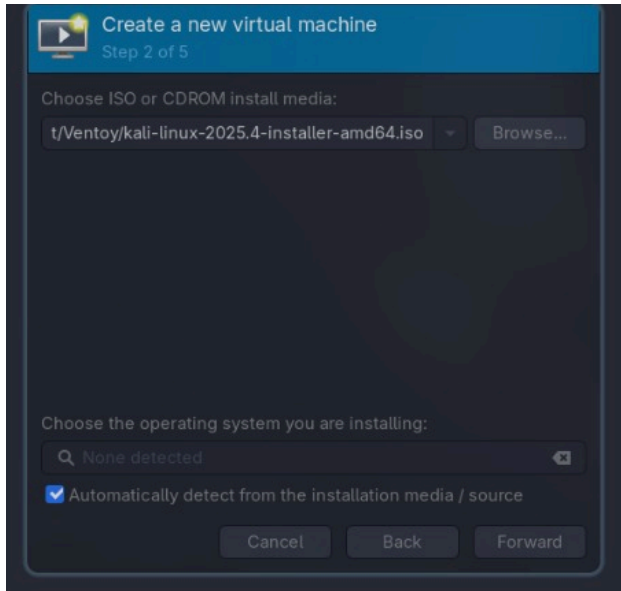
Step 2: Choose Installation Method

Select: Local install media (ISO image or CDROM)  
Click Forward



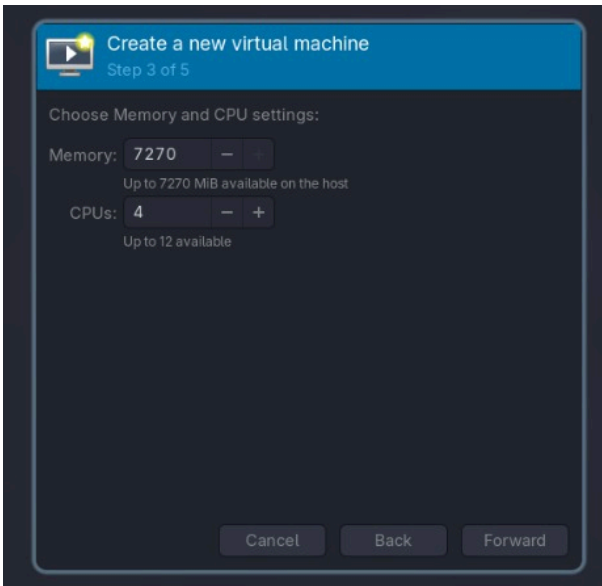
Step 3: Select ISO File

- Click Browse → Browse Local
- Select your OS ISO file (e.g., Ubuntu ISO)
- Click Forward



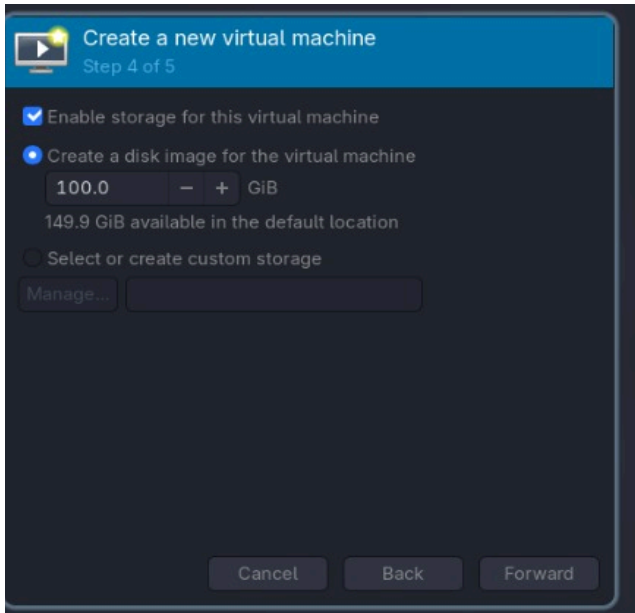
Step 4: Set Memory & CPU

RAM: 2048 MB (2 GB)  
CPU: 2 cores  
Click Forward



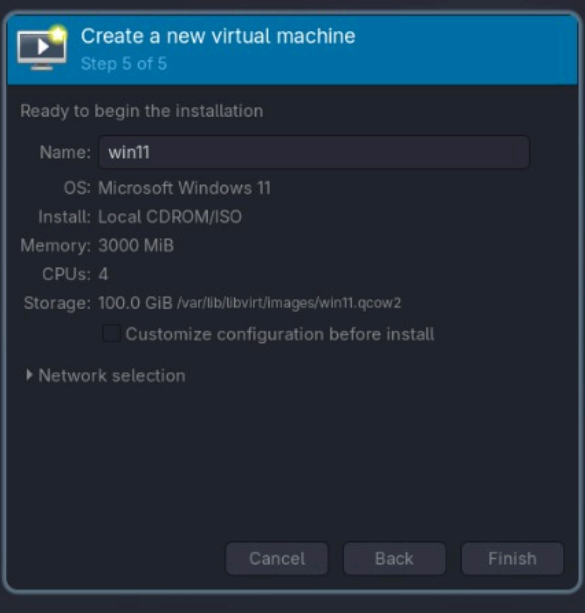
Step 5: Create Disk

- Select: Create a disk image
- Size: 20 GB
- Click Forward



Step 6: Final Configuration

- Name your VM (e.g., **Ubuntu-VM**)
- Ensure network is NAT (default)
- Click Finish



Step 7: Start Installation

- VM starts automatically
- OS installer screen appears

Step 8: Install Operating System

- Follow installation steps:
- Language
- Username

8. Result

KVM virtualization was successfully installed and configured, and a virtual machine was created and executed successfully.

9. Conclusion

This assignment demonstrated the installation and configuration of Kernel-based Virtual Machine and provided practical knowledge of virtualization and VM management.